

Cloud Computing and Software -- Guided Notes ANSWER KEY

Unit tp-3.5 | CompTIA Network+ N10-009 Obj. FC0-U71 3.5 | N10-008 FC0-U71 3.5

For instructor use / distribute after students complete the notes.

Question 1

SaaS (Software as a Service) delivers applications over the _____ through a web browser. The user does not install or maintain the software -- the _____ handles all updates and infrastructure.

Examples include _____, _____, and _____.

Answer: Internet; provider (vendor); Google Docs, Microsoft 365 Online, Salesforce (any three).

Question 2

Cloud storage services like _____, _____, and _____ let users store files on remote servers accessible from any device. Files are synchronized between devices and stored redundantly. The main risk is _____ if the account is compromised or the provider experiences an outage.

Answer: OneDrive, Google Drive, Dropbox (any three); data loss / inaccessibility.

Question 3

A _____ (virtual desktop) lets a user access a full computing environment hosted in the cloud through a thin client or browser. All processing and storage happen on the _____ server. This model simplifies _____ management because the desktop is in one centralized location.

Answer: Cloud desktop (VDI); remote (cloud); endpoint.

Question 4

API stands for _____ and allows different software applications to communicate with each other over a network. Cloud services expose _____ that developers use to integrate features (like payment processing, maps, or authentication) into their own applications. Today most public APIs use the _____ architectural style.

Answer: Application Programming Interface; APIs; REST.

Question 5

Web applications run inside a _____ and require no local installation. They use HTML, CSS, and _____ on the client side. Complex web apps may use frameworks like _____ or _____. A _____ (Progressive Web App) can be installed to the home screen of a mobile device and work offline.

Answer: Browser; JavaScript; React or Angular (examples); PWA.

Question 6

Cloud software updates are _____ (pushed automatically by the vendor). Users always have the _____ version without action. The downside is that businesses have _____ control over when updates occur. Enterprise cloud plans often offer _____ deployment rings to test updates before full

rollout.

Answer: Automatic; latest (current); less; staged.

Question 7

Multi-tenancy means multiple _____ share the same underlying cloud infrastructure, with their data logically _____ from each other. This reduces cost for the vendor through economies of scale.

Security concern: if the _____ is breached or misconfigured, another tenant's data could potentially be exposed.

Answer: Customers (tenants); separated (isolated); isolation layer.

Question 8 -- Real World Application

REAL WORLD: A company currently stores all customer data on a local file server. The IT manager wants to move to cloud storage. One employee raises a concern: 'What happens to our data if the cloud provider goes out of business or raises prices dramatically?' Explain the risk the employee is describing and two strategies to mitigate vendor lock-in.

Answer: The employee is describing vendor lock-in -- the risk that once data and workflows are tied to one cloud provider, it becomes difficult and expensive to switch. If the provider raises prices, experiences an outage, or shuts down, the company has limited ability to migrate quickly. Mitigation strategy 1 -- Multi-cloud: replicate or back up data to a second cloud provider (e.g., store primary data on Azure but maintain a backup copy on AWS S3). If one provider fails, the other has the data. Migration strategy 2 -- Use open standards: store data in standard formats (CSV, PDF, JPEG) rather than proprietary formats that only work with one vendor. Ensure the application supports data export at any time, so switching providers does not require reformatting data.